

Supplementary Materials

Nutritional Indices and All-Cause Mortality in GI Cancer: A Dual-Cohort NHANES Analysis

Tables S1–S7

Table S1: GI Cancer Site Distribution and Mortality Outcomes

Site	N	%	Deaths	PNI (mean±SD)	Median survival
Colon	175	55.9	100	49.8 ± 4.9	7.4 yr
Rectum	66	21.1	36	51.2 ± 5.8	9.2 yr
Liver	23	7.3	15	48.1 ± 6.2	5.8 yr
Stomach	19	6.1	10	50.5 ± 5.1	8.1 yr
Esophageal	15	4.8	7	47.9 ± 5.7	4.3 yr
Pancreatic	8	2.6	5	49.0 ± 4.8	6.0 yr
Gallbladder	6	1.9	3	51.3 ± 5.5	9.5 yr
Unknown GI	1	0.3	–	–	–
Total	313	100	169	50.8 ± 6.5	8.8 yr

Table S2: Full Baseline Characteristics by GI Tumor Status

Variable	GI Tumor	Non-Cancer	Other Cancer
Age, years	67.5 ± 13.8	45.5 ± 18.4	65.6 ± 15.0
Female (%)	64 (20.4)	7,172 (17.7)	1,876 (57.3)
Race/ethnicity (%)			
NH-White	160 (51.1)	14,887 (36.8)	2,131 (65.1)
NH-Black	45 (14.4)	5,909 (14.6)	375 (11.5)
Hispanic	101 (32.3)	16,394 (40.5)	659 (20.1)
Other	7 (2.2)	3,290 (8.1)	108 (3.3)
Education ≥ HS (%)	187 (59.7)	25,808 (63.8)	2,228 (68.1)
BMI, kg/m ²	28.5 ± 6.1	28.3 ± 6.6	28.4 ± 6.3
Albumin, g/dL	4.1 ± 0.4	4.2 ± 0.4	4.1 ± 0.4
Lymphocyte, μL^{-1}	1,972 ± 990	2,209 ± 719	1,989 ± 790
Cholesterol, mg/dL	195 ± 44	196 ± 43	199 ± 43
PNI	50.8 ± 6.5	53.2 ± 5.2	51.4 ± 6.0
CONUT	1.1 ± 1.4	0.7 ± 1.0	1.0 ± 1.2
GNRI	115.0 ± 12.2	116.4 ± 12.4	115.8 ± 12.0
Deaths (%)	169 (54.0)	7,659 (18.9)	1,042 (31.8)
Median f/up	8.8 yr	13.3 yr	10.5 yr

Table S3: Baseline Characteristics by PNI Tertile Among GI Cancer Patients

Variable	Low PNI (T1)	Mid PNI (T2)	High PNI (T3)
N	105	104	104
Age, years	72.6 ± 9.4	66.4 ± 15.5	63.6 ± 13.8
Female (%)	15 (14.3)	26 (25.0)	23 (22.1)
Non-Hispanic White (%)	52 (49.5)	57 (54.8)	51 (49.0)
Education ≥ high school (%)	64 (61.0)	58 (55.8)	65 (62.5)
PNI	44.8 ± 3.5	50.5 ± 1.4	57.5 ± 4.1
CONUT	2.1 ± 1.7	0.9 ± 0.9	0.3 ± 0.5
GNRI	110.2 ± 12.6	115.5 ± 10.4	119.5 ± 11.7
Albumin, g/dL	3.8 ± 0.4	4.1 ± 0.3	4.4 ± 0.3
Lymphocyte, μL^{-1}	1,376 ± 389	1,820 ± 516	2,739 ± 1,049
Total cholesterol, mg/dL	184 ± 45	196 ± 41	205 ± 43
BMI, kg/m^2	28.3 ± 6.1	28.5 ± 5.6	28.7 ± 6.6
Deaths (%)	74 (70.5)	48 (46.2)	47 (45.2)
Median survival	5.5 yr	7.8 yr	13.9 yr

Table S4: Cohort-Stratified Analysis

Index	Cohort	N	Events	HR (95% CI) per 1-SD	p
PNI	NHANES III	88	68	0.69 (0.51–0.93)	0.015
PNI	Continuous NHANES	225	101	0.75 (0.56–1.00)	0.047
CONUT	NHANES III	88	68	0.72 (0.56–0.93)	0.011
CONUT	Continuous NHANES	225	101	0.69 (0.57–0.85)	<0.001
GNRI	NHANES III	88	68	0.64 (0.48–0.84)	0.001
GNRI	Continuous NHANES	225	101	0.68 (0.55–0.84)	<0.001

Cohort-by-index interaction p-values: PNI p=0.480, CONUT p=0.940, GNRI p=0.950.

Table S5: Extended Covariate Sensitivity Analysis

Model	Index	N	Events	HR (95% CI) per 1-SD	p
Basic (age+sex+race)	PNI	313	169	0.78 (0.63–0.96)	0.020
+ Education + PIR	PNI	313	169	0.76 (0.62–0.94)	0.012
+ Smoking + Comorbidity [†]	PNI	133	61	0.68 (0.48–0.97)	0.035
Basic (age+sex+race)	CONUT	313	169	0.71 (0.61–0.83)	<0.001
+ Education + PIR	CONUT	313	169	0.71 (0.60–0.82)	<0.001
+ Smoking + Comorbidity [†]	CONUT	133	61	0.63 (0.50–0.80)	<0.001
Basic (age+sex+race)	GNRI	313	169	0.67 (0.57–0.80)	<0.001
+ Education + PIR	GNRI	313	169	0.66 (0.56–0.79)	<0.001
+ Smoking + Comorbidity [†]	GNRI	133	61	0.72 (0.55–0.93)	0.013

PIR: income-to-poverty ratio. [†]Smoking and comorbidity data only available in Continuous NHANES 2005–2016 subset.

Table S6: Survey-Weighted Cox Regression (2005–2016)

Index	N	Events	HR (95% CI) per 1-SD	p
PNI	270	134	0.87 (0.61–1.23)	0.432
CONUT	270	134	0.76 (0.63–0.91)	0.003
GNRI	270	134	0.73 (0.59–0.90)	0.003

Survey weights: WTMEC2YR/(n cycles). Adjusted for age, sex, race/ethnicity.

Table S7: Joint PNI×NLR, CRP Interaction, and CALLY Index**S7a: Joint PNI × NLR Analysis (n=258, events=125)**

Group	HR (95% CI)	p
High PNI + Low NLR (ref)	1.00	–
Low PNI + Low NLR	1.03 (0.61–1.75)	0.902
High PNI + High NLR	0.79 (0.41–1.54)	0.492
Low PNI + High NLR	1.72 (1.06–2.63)	0.028

RERI = -0.97 (95% CI -2.61 to -0.10), indicating overlapping rather than synergistic pathways.

S7b: CRP Interaction (n=250, events=120)

Predictor	HR (95% CI)	p
PNI per 1-SD	0.84 (0.63–1.10)	0.205
log(CRP) per 1-SD	1.18 (0.92–1.50)	0.186
PNI × CRP interaction	–	0.780

S7c: CALLY Index Comparison (n=250, events=120)

Index	HR (95% CI)	p	C-statistic
PNI	0.85 (0.67–1.09)	0.200	0.667
CALLY	0.85 (0.67–1.09)	0.200	0.659

CALLY = albumin × lymphocyte / CRP. Neither index reached statistical significance in this subset with CRP data. C-statistic difference was negligible ($\Delta=0.008$), indicating no advantage of CALLY over PNI.